VAJRA

Operator's User Manual
Visual Audio Joint-modal Rendering Architecture

Applies to	VAJRA 2.5
Audience	The operator at the interface. No programming assumed.
Companion documents	Deployment & Operations Guide (installing and running) · Technical Writeup (how the models work)
Scope	Every tab, every control, recommended values, and how to read the quality reports

This manual explains how to **use** VAJRA. It does not cover installation — that is the Deployment Guide — and it does not explain the models' internal mechanisms, which is the Technical Writeup. Read section 1 and 2 once, then use the module sections as reference.

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1. Before You Start

Starting the system

```
bash run.sh
```

Then open **http://localhost:7860** in a browser on the same machine. If the machine is remote and a link was enabled at start-up, use the printed **.gradio.live* address instead.

Caution. A public link gives anyone who has it full use of the system, with no password. Only enable it if you intend that, and never on a machine holding sensitive material.

Five things to know before your first job

- **One job at a time.** The system deliberately runs one GPU task at a time. If you press a second button, it waits rather than competing for memory. This is normal — nothing has hung.
- **The first run of any module is slow.** Models load on first use, and some are several gigabytes. Later runs in the same session are much faster.
- **Every result is saved automatically** to the `outputs/` folder, named `vajra_<what>_<timestamp>`. You do not need to download it to keep it.
- **Some modules may not be installed.** The system installs selectively. If a module reports that its environment is missing, that is a setup choice, not a fault — the message names what to add.
- **Nothing leaves the machine.** There is no cloud service behind any module. All processing is local.

Choosing your source material

Result quality depends more on the input than on any setting. For work involving a person's face or voice:

Property	Aim for	Why
Framing	Front-facing, whole face visible	Profiles and partial faces degrade every face and lip-sync model
Lighting	Even, no harsh shadow across the face	Shadow edges confuse face detection and appear as artefacts
Audio	Clean speech, little background noise	The voice reference is taken from your recording; noise is cloned too
People in frame	One speaker	The lip-sync models cannot tell which mouth to drive
Length	1–3 minutes for a first trial	Long clips work but take proportionally longer to verify

2. The Interface

Header and status strip

Element	Meaning
Quick-access chips (top right)	Jump to a module
Light / dark toggle	Both themes are fully supported; use whichever is comfortable
ONLINE · 100% OFFLINE / LOCAL	The interface is running, and no external service is in use. Both statements are true at once.
Compute	The GPU detected, or “CPU (no GPU)” — in which case only Media Studio will work
Modules	How many modules this build exposes

Element	Meaning
Signals strip	A decorative activity indicator. It is **not** connected to your audio and shows nothing about the current job.

Conventions used in every module

- **Left column is input, right column is output.**
- **The primary action is the amber button.** Secondary steps are outlined.
- **Advanced settings are inside collapsible panels.** The defaults are chosen to be correct for most work; open the panel only when you have a reason.
- **Status appears beneath the output** — a tick for success, a warning triangle for something you must fix, a cross for a failure.

Tip. If a result looks wrong, your first move should be to read the status line and open any report panel underneath it. The system usually states what it did and why.

3. Module 01 — Video Edit

What it does. Takes a talking-head video, transcribes it, lets you change the words, then re-voices only what you changed in the speaker's own voice and re-synchronises their lips to match.

What makes it different from re-recording. Everything you do not change is left exactly as filmed — the audio outside an edit is untouched to the last sample, and only the video frames spanning an edit are re-rendered.

Step-by-step

1. **Upload a talking-head video** in the left column.
2. **Set Spoken language** if you know it. Leave on *Auto-detect* only if you are unsure — see the caution below.
3. Press **Extract Transcript**. This takes up to a minute or two. The transcript appears in two panes: the left is a read-only reference, the right is yours to edit.
4. **Edit the right-hand pane.** Change words; keep one segment per line and do not merge or delete lines.
5. Press **Apply Edits & Re-sync**. Progress is reported as it works.
6. **Review the result:** the video appears top right, with the original and edited audio below it for comparison.
7. Open **Routing & detectability report** to see how the edit was made and how convincing it measures. See §10.

Caution. Language auto-detection reads only the first half-minute and can mistake accented English for a related language — after which it writes English words in that language's script. If the transcript comes back in the wrong script, set **Spoken language** explicitly and extract again.

Main controls

Control	Choices / range	Default	What it does
Spoken language	Auto-detect, or one of 18	Auto-detect	Tells the recogniser what language is being spoken
Lip-sync	Best quality · Fast	Best quality	Best quality regenerates the mouth region; Fast is more robust on difficult footage at lower resolution
Diffusion steps	10 – 50	20	Refinement passes for the mouth. Higher is slower with diminishing return; 20 is a good balance
Guidance	1.0 – 3.0	1.5	How strongly the audio drives the mouth. Raise slightly if sync looks loose; too high looks exaggerated

Advanced panel — "Realism & routing"

These defaults are deliberate. Change one at a time so you can tell what it did.

Control	Default	When to change it
Match the splice to its surroundings	On	Leave on. Turning it off skips loudness, tone, room and noise matching — useful only to hear what the matching is doing
Word-level edits	On	Leave on. Off regenerates a whole transcript line for any change, which discards audio that could have stayed genuine
Re-sync only edited windows	On	Leave on. Off re-renders every frame of the video, which is far slower and touches footage that had no reason to change
Separate background audio	Never	Set to *Auto* or *Always* only when there is music or significant ambience to protect. It is a lossy round trip
Max time-stretch	0.15	How far replacement speech may be squeezed or stretched to fit. Above about 0.15 stretching becomes audible; lower it for critical work
Replace the entire track	Off	Turn on only when you are changing the whole script — see below

Control	Default	When to change it
Quality tier	Auto (best available)	Leave on Auto. Select a specific tier only to compare methods; a tier your language or installation cannot deliver falls back and says so
Allow non-commercial engines	Off	Only if your use permits a non-commercial licence

How much can you change?

This is the most important limitation to understand. Replacement audio must occupy the same time as the words it replaces, because the soundtrack has to stay in step with the picture.

If your new wording is…	What happens	What to do
about the same length	Fits directly. Best results.	Nothing
slightly longer or shorter	Absorbs neighbouring silence, or is gently stretched within the limit	Nothing
noticeably longer	The whole line is re-voiced rather than cut short. Reported in the panel.	Acceptable, but more of the line becomes synthetic
much longer (e.g. twice)	Cannot fit. You will see “duration capped”.	Reword closer to the original length, or use **Replace the entire track**

Tip. Change **a few words per line**, not whole lines, whenever the meaning allows. That is where this system's quality lives: a two-word change keeps most of the real recording, and real recording never needs disguising.

Replace the entire track

Use when the whole script changes. Every line is re-voiced at a natural pace and the picture is then re-timed to match the new length.

Caution. Two consequences. **None of the original audio survives** — there is nothing left to blend into. And the video is stretched or compressed to fit; beyond roughly 25% this is visible, because gestures and blinks run slow or fast. The report states the factor and warns when it crosses that point.

4. Module 02 — Voice Generation

What it does. Audio-only work, in two sub-tabs. **Voice Edit** changes words inside an existing recording. **Voice Clone** speaks new text in a voice taken from a reference clip.

Note. This module is **English only** in this build, and requires its own environment and weights. Press **Check service status** first. Its first start loads three models and takes several minutes — this is normal.

Voice Edit

1. Upload the **audio to edit**.
2. Press **Extract transcript**. Do not type it by hand unless you must — a typed transcript that does not match the audio will misplace the edit.
3. The edited box is filled with the same text. **Change only the words you want re-spoken.**
4. Press **Edit the speech**. The result appears on the right.

Control	Choices / range	Default	What it does
Alignment granularity	Word (precise) · Segment (safer)	Word	How finely the text is matched to the audio. Use Segment if word alignment misplaces the edit
Mask expansion	0.00 – 0.50	0.10	How far beyond the changed words the model may regenerate. Higher blends the join better but keeps less of the original recording

Voice Clone

1. Upload a **reference clip** — a few seconds of clean speech from the target speaker. Quality here matters more than anything else on the tab.
2. Type the **text to speak**.
3. Press **Generate speech**.

Control	Range	Default	What it does
Emotion guidance	0.0 – 5.0	1.0	How strongly expressive cues in the text are followed. Raise for more expression; too high sounds theatrical
Voice guidance	0.0 – 5.0	1.0	How closely the output holds to the reference voice. Raise if the clone drifts; too high can sound strained

Tip. A clean six-second reference beats a noisy thirty-second one. The system copies whatever is in the reference, including room echo and background noise.

5. Module 03 — Image Generation

What it does. Produces a photoreal image from a written description.

1. Write a **prompt** — describe subject, setting, lighting and framing.
2. Optionally add a **negative prompt** — things to avoid.
3. Set size and quality if needed, then press **Generate**.

Control	Range	Default	Guidance
Model	Photoreal · General purpose	Photoreal	Photoreal for people and scenes; General purpose for illustration or unusual subjects
Width / Height	512 – 1536, steps of 64	1024 × 1024	Stay near the default. Extreme aspect ratios produce distorted anatomy

Control	Range	Default	Guidance
Steps	4 – 50	30	Refinement passes. Below ~20 looks unfinished; above ~35 rarely helps
Guidance	0.0 – 9.0	6.0	How strictly the prompt is followed. Low drifts off-prompt; high looks over-saturated and harsh
Seed	–1 for random, or any number	–1	Set a number to reproduce an image exactly, or to change one setting while holding composition

Tip. To refine an image you like: note its seed, keep it fixed, and change one thing at a time. With a random seed every change alters the whole picture and you learn nothing.

6. Module 04 — Face Swap

What it does. Takes the identity from a source face and applies it to a target image, or to every frame of a target video.

1. Upload the **source face** — the identity to copy FROM. A clear, front-facing photo.
2. Choose **Target type**: Image or Video.
3. Upload the **target** — the picture or footage to paste INTO.
4. Choose a **face enhancer**, then press **Swap Face**.

Control	Choices	Default	What it does
Target type	Image · Video	Image	Video processes every frame and re-attaches the original audio
Face enhancer	Standard · High detail · None	Standard	Restores detail the swap cannot produce. **High detail** is sharper but image-only and non-commercially licensed. **None** leaves a visibly soft face

Note. Enhancement is not cosmetic. The swap works on a small aligned crop, so its output has a resolution ceiling; without restoration the face reads soft against a sharp scene. Leave it on **Standard** unless you have a reason.

7. Module 05 — Text → Video

What it does. Two related jobs. With a prompt alone it generates a short video with its own synchronised audio. With a reference photo it animates that photo, preserving the person's identity.

1. Write a **prompt** describing the action.
2. **Optionally** add a reference photo to animate instead of generating from scratch — and pick the motion engine.
3. Set resolution, duration and frame rate. Press **Generate**.

Control	Range	Default	Guidance
Reference photo	optional	none	With a photo, the identity is preserved and the photo becomes the opening frame
Motion-video engine	Identity-preserving · Synchronized audio	Identity-preserving	Only used when a photo is supplied. Identity-preserving is better for people; the other is faster and adds a soundtrack
Resolution	480p and above	480p (832×480)	Raise only if you have VRAM to spare — cost climbs steeply
Duration	1 – 4 s	3 s	These are short clips by design. Longer means much slower
Frame rate	25 – 60	25	Higher is smoother but generates proportionally more frames
Steps	10 – 50	40	Refinement passes for the audio engine
Guidance	1.0 – 5.0	1.0	Prompt adherence for the identity engine

Caution. This is the heaviest module. Both engines are very large downloads and need substantial VRAM. If it is not installed, that was a deliberate setup choice — see the Deployment Guide.

8. Module 06 — Image Edit

What it does. Changes an existing image according to a written instruction, optionally combining up to three reference images.

1. Upload **one to three images**.
2. Write an **edit instruction** in plain language — for example, “place the product from image 2 onto the desk in image 1”.
3. Press **Edit**.

Control	Range	Default	Guidance
Images	1 – 3	—	With several, refer to them by number in the instruction
Edit instruction	free text	—	Say what should change. Instructions work better than descriptions of the desired final state
Steps	10 – 50	40	Refinement passes
Guidance	1.0 – 8.0	4.0	How literally the instruction is followed. Too high distorts the untouched parts of the image
Seed	–1 or a number	–1	Fix to reproduce a result

9. Module 07 — Media Studio

What it does. Preparation and conversion tools that use no GPU at all. Use these to get material into the right shape before spending GPU time on it.

Tool	Inputs	Use it to
Trim video / audio	File, start and end in seconds	Cut to the section you actually need — the single most effective way to reduce processing time
Grab frame	Video, time in seconds	Take a still to use as a face-swap source or a video reference photo
Clean audio	Audio or video	Reduce background noise before cloning a voice from it
Change background	Portrait, colour choice	Replace a portrait's background with white, green, grey or black
Burn captions	Video	Write subtitles permanently into the picture
Convert	Any file; MP4 / compress / MP3 / WAV	Fix an unusable format, or shrink a file
Resize	Video, width and height	Match a required output size
GIF	Video, fps 5–24, width 240–720	Produce a short animated GIF
Merge	Two or more files, video or audio	Join clips end to end
Session outputs	—	Collect everything produced this session as a single zip

Tip. Look for **"→ Send to"** buttons. They push a result straight into another module without saving and re-uploading — for example a grabbed frame into Face Swap, or a trimmed clip into Video Edit.

10. Reading the Quality Report

After a Video Edit, open **Routing & detectability report**. It has up to four panels. This is the part of the system that tells you whether to trust the result, so it is worth understanding.

Routing — how the edit was made

Field	Meaning
Tier	The method used. **A** regenerates only the changed words using the surrounding audio — best. **C** regenerates a pause-bounded phrase. **D** regenerates a whole line.
Method	Whether the edit was in-place infill or phrase regeneration
Language	The language used for synthesis
Ceiling	A better tier is possible for this language but is not installed. Worth reporting to whoever maintains the system.
Note	Anything that limited the result, in plain words

Detectability scorecard — how convincing it is

The headline is the **seam percentile**. The system finds every genuine word boundary in your own recording, measures the join across each, and reports where your edit sits in that distribution.

Reading	Interpretation	Action
under 50th percentile — *inaudible*	Smoother than most real word boundaries in this recording	None
50th–80th — *unlikely to be noticed*	Within the normal range for this speaker	None
80th–95th — *possibly audible*	More abrupt than most genuine boundaries	Listen carefully; consider rewording
above 95th — *likely audible*	More abrupt than nearly every genuine boundary	Reword, or move the edit to a natural pause

Alongside it: **loudness** (difference from the surrounding audio — under 1 dB is imperceptible), **tone** (spectral difference — under 4 dB is good), and **floor** (whether the background hiss continues across the edit).

Caution. What the score does not tell you. It measures the *join*, not the quality of the replacement. A perfectly blended splice of poor synthesis still shows PASS. The report lists four things it does not measure — speaker similarity, naturalness, intelligibility and lip-sync accuracy. **Always listen to the result yourself.**

Lip-sync panel

Reports how many regions of video were re-rendered and what percentage of the clip that was — the rest is your original footage, untouched. If it says the whole video was re-synced, it also says why. **Texture match** reports how many frames had their grain and sharpness matched to the surrounding face.

Whole-track retake panel

Appears only when you used **Replace the entire track**. States the original and new lengths, the factor by which the picture was re-timed, and a warning if that factor is large enough to see.

11. Recommended Settings

Starting points that work. Change one thing at a time from here.

Task	Settings
Video Edit — first attempt	All defaults. Change two or three words in a 1–2 minute clip. Set the spoken language explicitly.
Video Edit — critical work	Diffusion steps 30; Max time-stretch 0.10; keep every realism option on; reword to match the original syllable count
Video Edit — quick check	Lip-sync *Fast*; steps 10. Use to confirm the wording before committing to a full render
Video Edit — music or ambience present	Separate background audio → *Always*
Voice Clone — best fidelity	6–10 seconds of clean reference; both guidance sliders at 1.0; raise voice guidance only if the clone drifts
Image Generation — photoreal person	Photoreal model; 1024×1024; steps 30; guidance 6.0; negative prompt listing artefacts you keep seeing
Face Swap — video	Front-facing source; enhancer *Standard*; trim the clip first in Media Studio
Text → Video — first attempt	480p; 3 seconds; 25 fps. Raise only after you have a result you like

12. Common Tasks

Change what a person says on camera

1. **Media Studio** → **Trim video**: cut to the section containing the sentence you want to change.
2. **Video Edit**: upload the trimmed clip, set the spoken language, Extract Transcript.
3. Edit only the words that must change; keep the length similar.
4. Apply Edits & Re-sync.
5. Compare **Original audio** and **Edited audio** side by side, then watch the video.
6. Open the report. If the seam percentile is high, reword and try again.

Make a person say something entirely different

1. As above, but tick **Replace the entire track** in the advanced panel.
2. Aim for a script that takes roughly the same time to speak as the original.
3. Check the retake panel for the re-timing factor. Above about 1.25 or below 0.8, the picture will visibly run slow or fast — shorten or lengthen the script.

Put someone's face into an existing photograph

1. **Media Studio** → **Grab frame** if your source is a video.
2. **Face Swap**: source = the face to copy, target type = Image, target = the photograph.
3. Enhancer *Standard*. Swap Face.

Produce a spoken clip in someone's voice

1. **Media Studio** → **Trim audio**, then **Clean audio**, to get a few seconds of clear speech.
2. **Voice Generation** → **Voice Clone**: upload it as the reference, type the text, Generate.

Reproduce an image you generated before

1. Find the seed in the status line of the original run, or in the report.
2. Enter the same prompt, the same seed, and the same size and steps. The result will be identical.

13. When Something Goes Wrong

What you see	What it means	What to do
“Upload a video first” / “Transcript is empty”	A required input is missing	Supply it
“… is not installed — missing the environment / the weights”	That module was not installed on this machine	Not a fault. Ask whoever set the system up to add that module
Transcript in the wrong language or script	Auto-detection chose wrongly and transliterated	Set **Spoken language** explicitly and extract again
“Duration capped … off by N seconds”	Your new wording needs far more time than the audio it replaces	Reword shorter, or use **Replace the entire track**
Report says a whole line was re-voiced	The change was too extensive to localise	Normal. Change fewer words per line to keep more of the original
Voice Generation: “503” or connection refused	Its services had not finished starting	Wait and retry — the first start takes several minutes. Use **Check service status**

What you see	What it means	What to do
Edited voice sounds processed or noisy	Usually over-correction, or a poor voice reference	Check the reference is clean. As a test, turn off **Match the splice to its surroundings** and compare
Mouth looks soft or blurred	The generated region is lower resolution than the face	Report it — texture matching should be handling this, and the report says whether it ran
Nothing happens after pressing a button	Another GPU job is running; yours is queued	Wait. This is by design, not a hang
The interface is unreachable	The application is not running	Restart with `bash run.sh`

14. Limits and Good Practice

Know these before you promise a result

- **Editing cannot add much speech.** Replacement audio must fit the time it replaces. Substantially longer scripts require re-timing the picture, which is visible beyond about 25%.
- **Voice Generation is English only** in this build. Other languages fall back to a lower tier, and the report says so.
- **One speaker per edit.** The lip-sync models cannot tell which mouth belongs to which voice.
- **Clean, front-facing footage works; difficult footage degrades.** Profiles, occlusion and poor light all reduce quality.
- **The quality score measures joins, not content.** Always watch and listen to the result yourself before relying on it.
- **Video generation produces short clips** — a few seconds — and is the most demanding module by a wide margin.

Working habits that pay off

- **Trim first.** Processing time scales with length; most jobs need only a section.
- **Change one setting at a time.** Otherwise you cannot attribute a difference to anything.
- **Keep a note of seeds** for images you may need to reproduce.
- **Read the report every time.** It is quicker than re-listening, and it tells you which stage to adjust.
- **Keep your source material.** Every result is reproducible from the input, the settings and the seed — but not from the output.

Responsible use

This system produces synthetic media that can be mistaken for a genuine recording. It is intended for authorised research and capability development in a controlled environment. Keep a record of what you generated and why; do not represent output as a genuine recording of any person; and follow your organisation's authorisation and record-keeping rules. The system is deliberately offline so that material and results stay on the machine.

Where to Look Next

Question	Document
How do I install or move this to another machine?	Deployment & Operations Guide
How does a given model actually work?	Technical Writeup
What is the architecture?	Architecture Diagrams (with speaker notes)

Question	Document
How large is each model, and what does a module cost on disk?	Model Footprint reference
What is verified automatically, and what is not?	Test documentation